

IBSIMU Beam Extraction Report

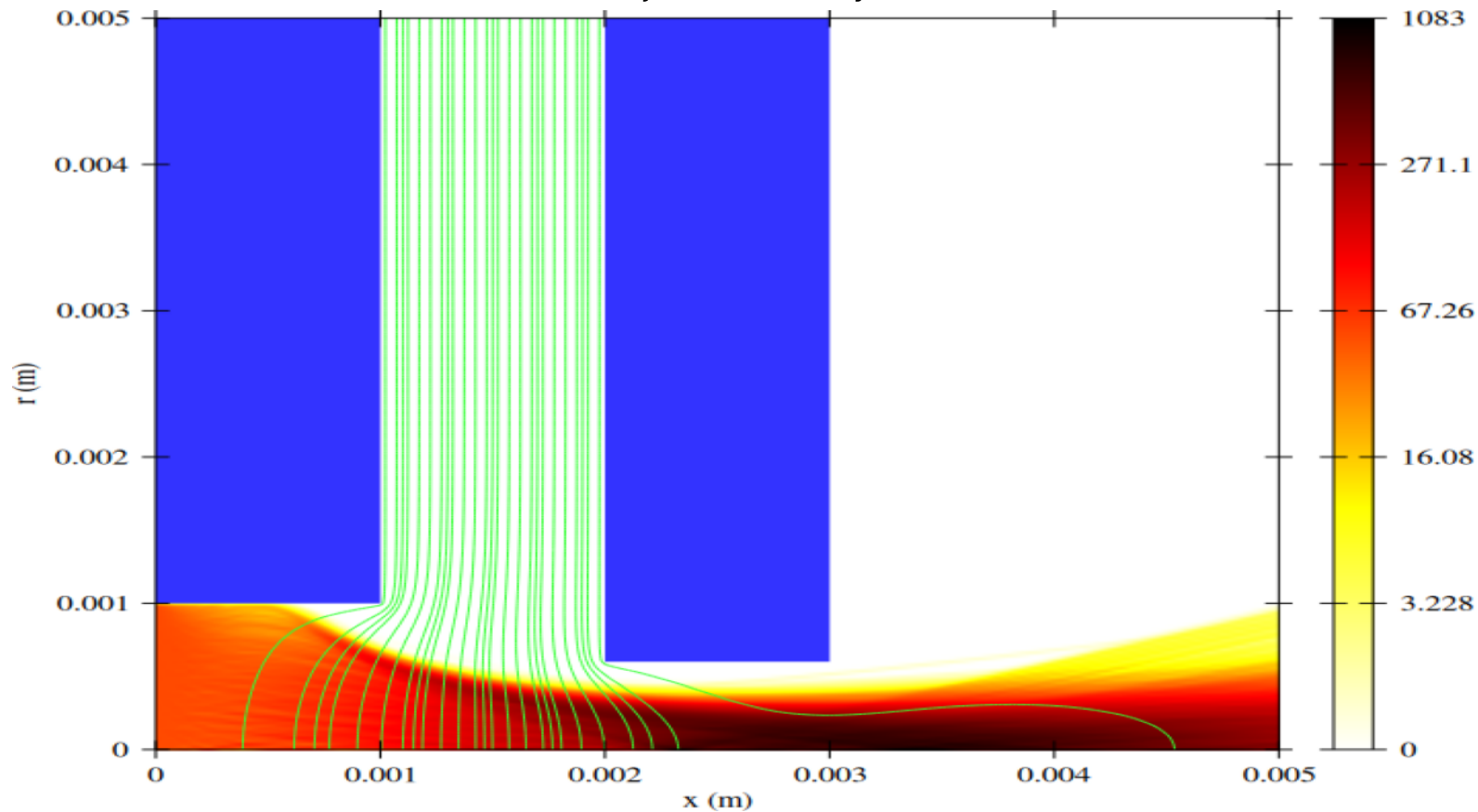
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--- Simulation Parameters ---
Grid points: 241
Particles: 5000
Current density: 600 A/m²
Beam energy: 5.0 eV
Species: Xe⁺
Mass: 131.29 amu, Charge: 1.0 e
Solver: BiCGSTAB
Beam mode: Energy/Temp
B-field: None

--- Electrodes ---
#      Gap      Apt    V(kV)    Thk    Chm    Wall
1       0.0       1     0.0       1      0      0
2       1.0      0.6    -8.0       1      0      0

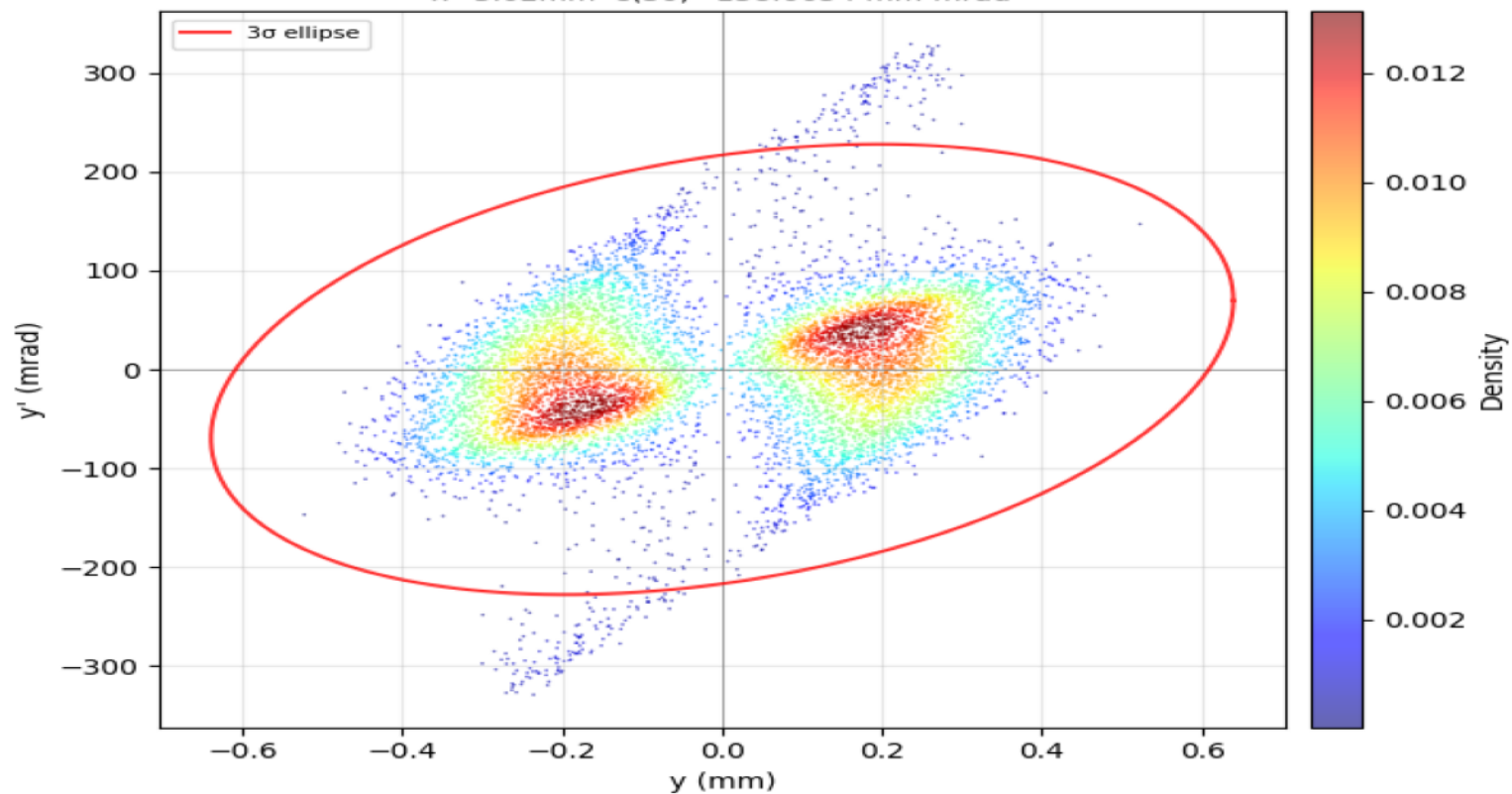
--- Results at Electrode Exit ---
Exit position: 3.00 mm
Divergence: 92.013 mrad (5.272°)
RMS radius: 0.2019 mm
Beam current: 1.2381e-04 A
Grid interception: 0.0%
Transmission: 100.0%
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Particle Trajectories (Axisymmetric)

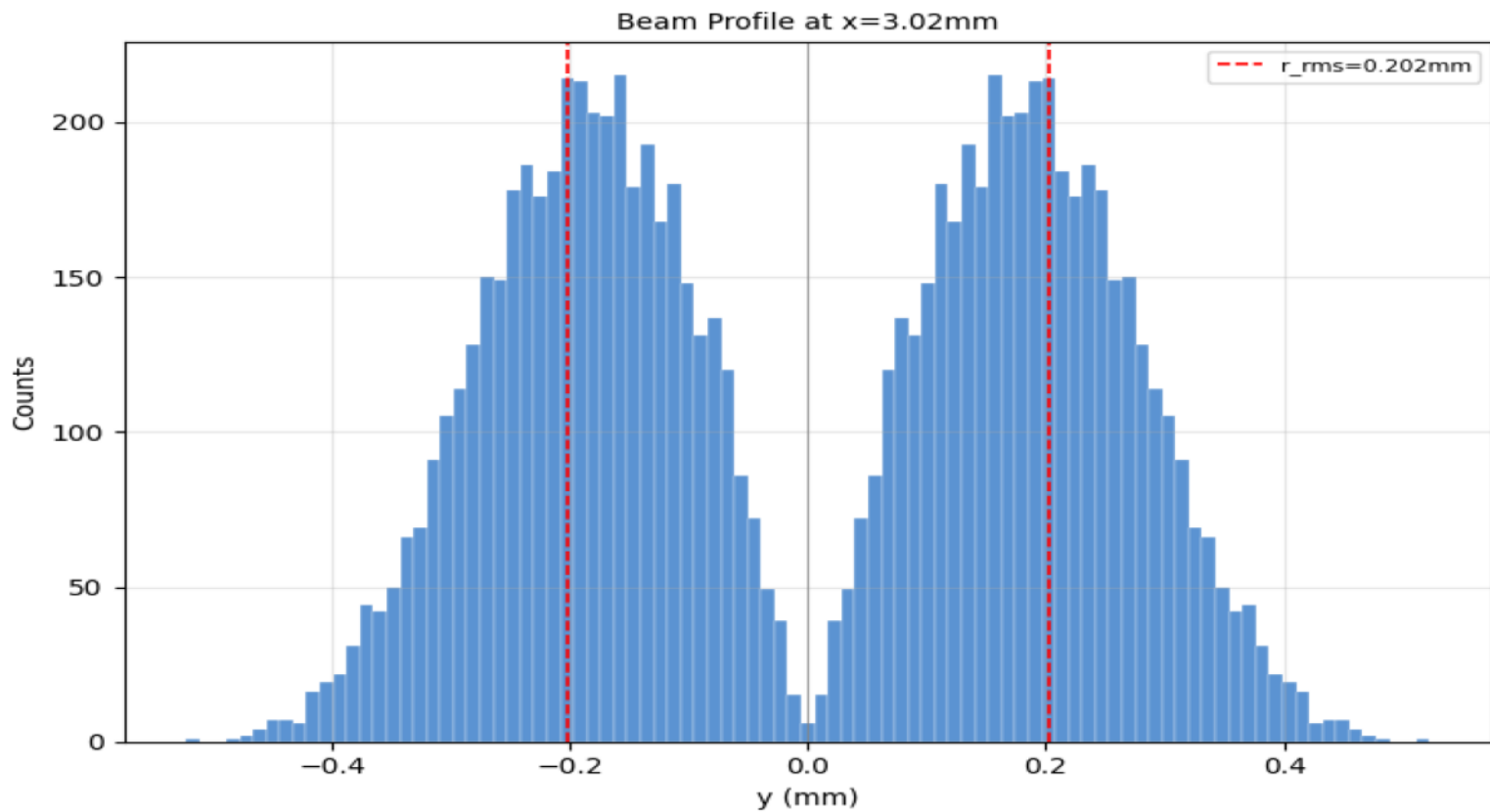


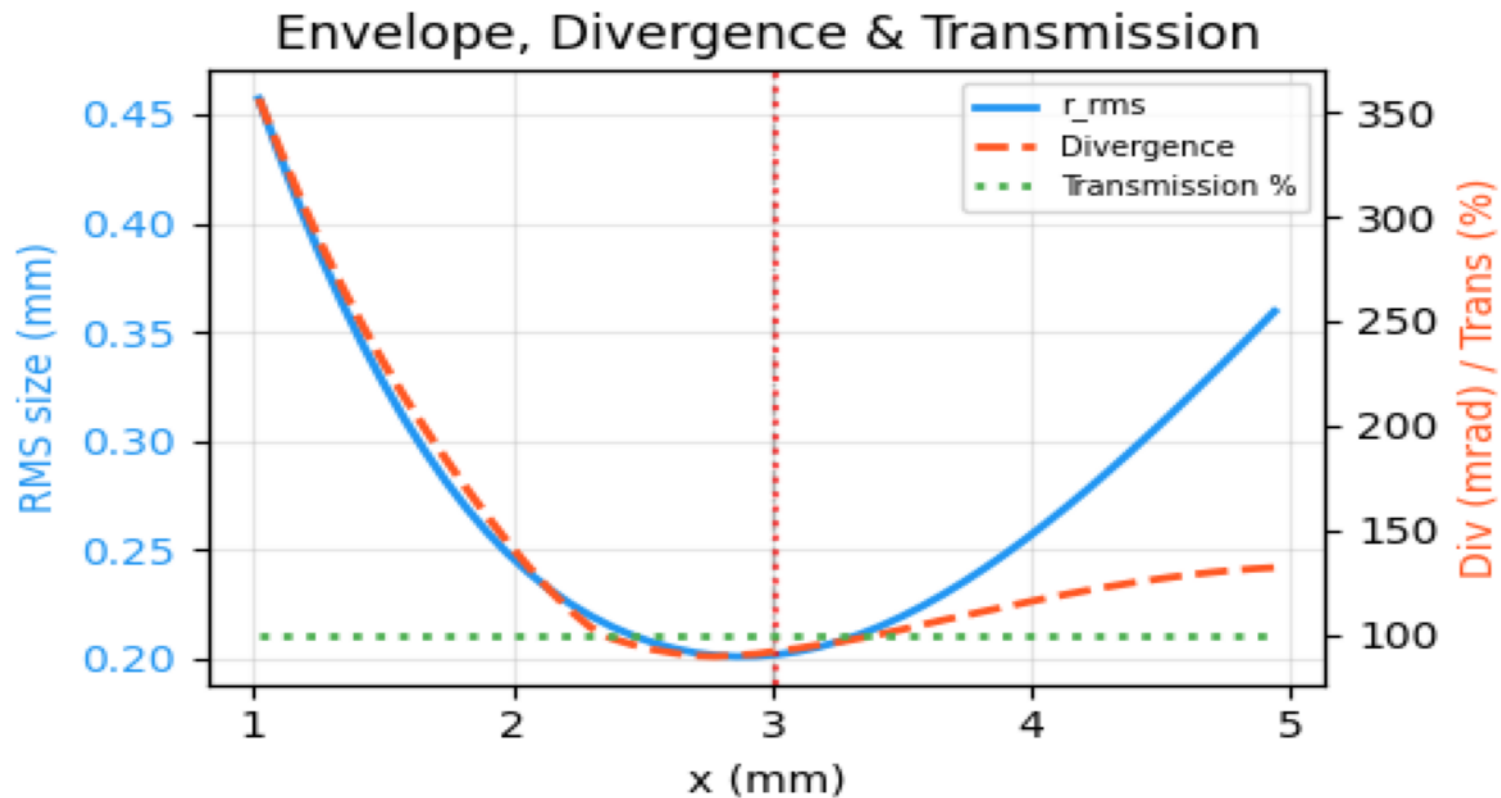
Phase Space

$x=3.02\text{mm}$ $\epsilon(3\sigma)=138.6654\text{ mm-mrad}$



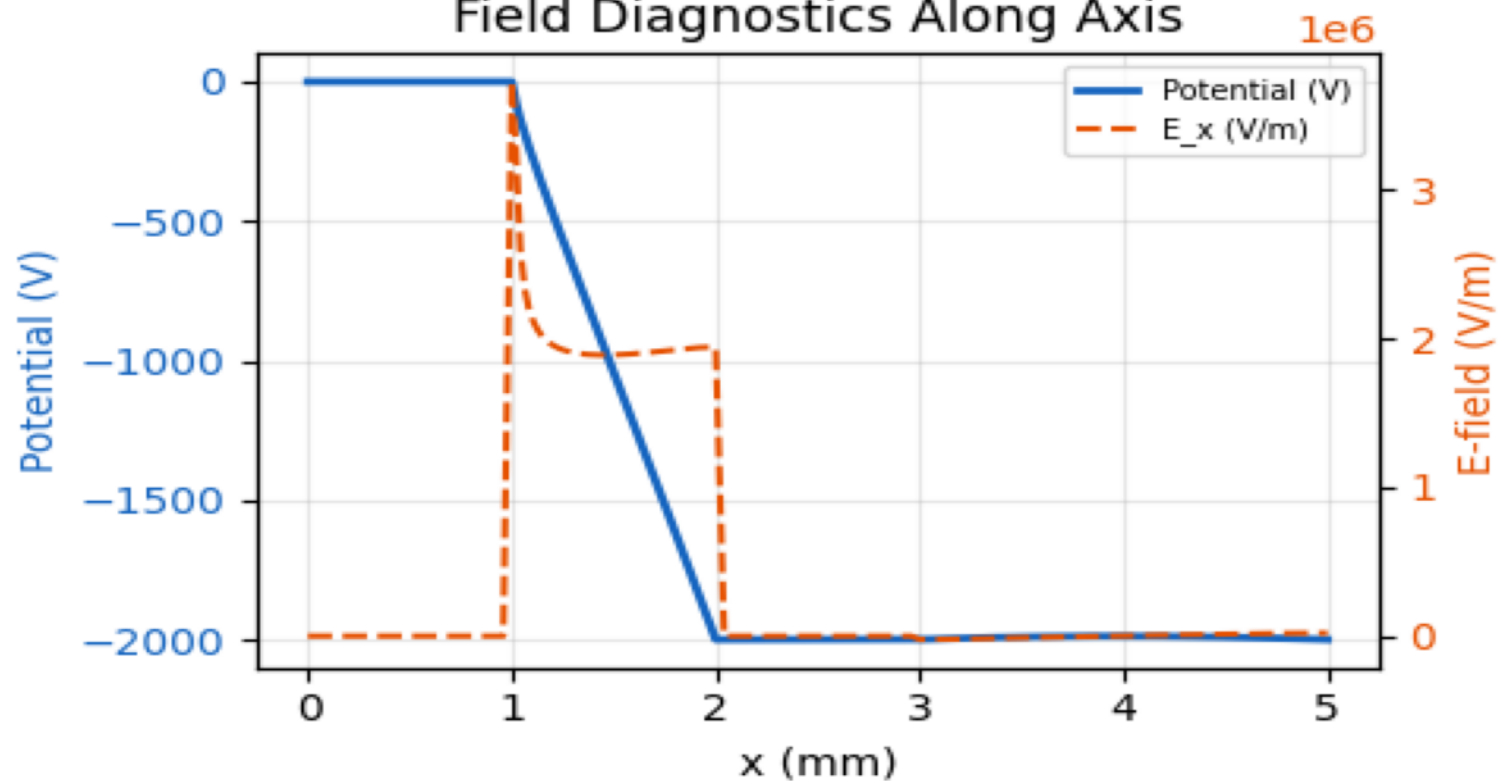
Beam Profile





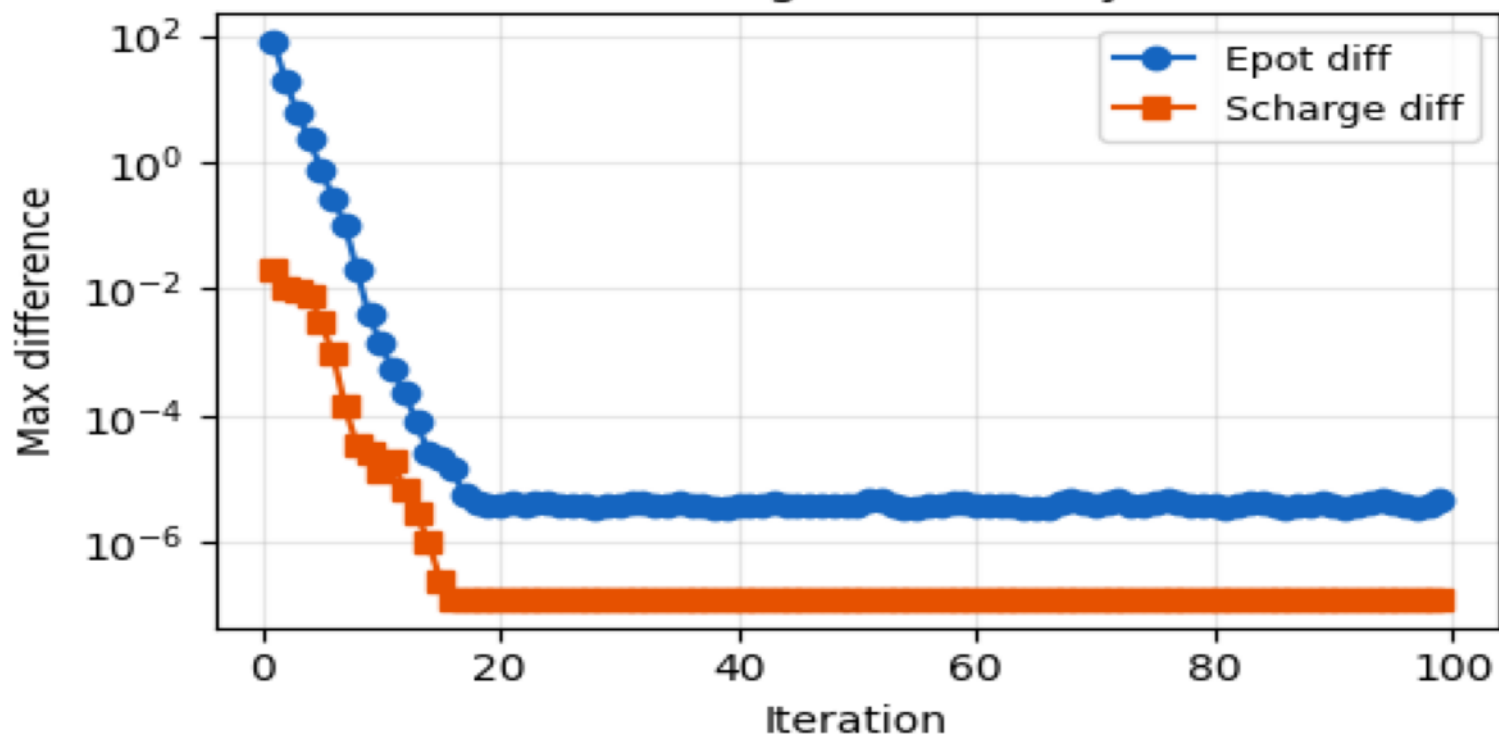
Field Diagnostics

Field Diagnostics Along Axis



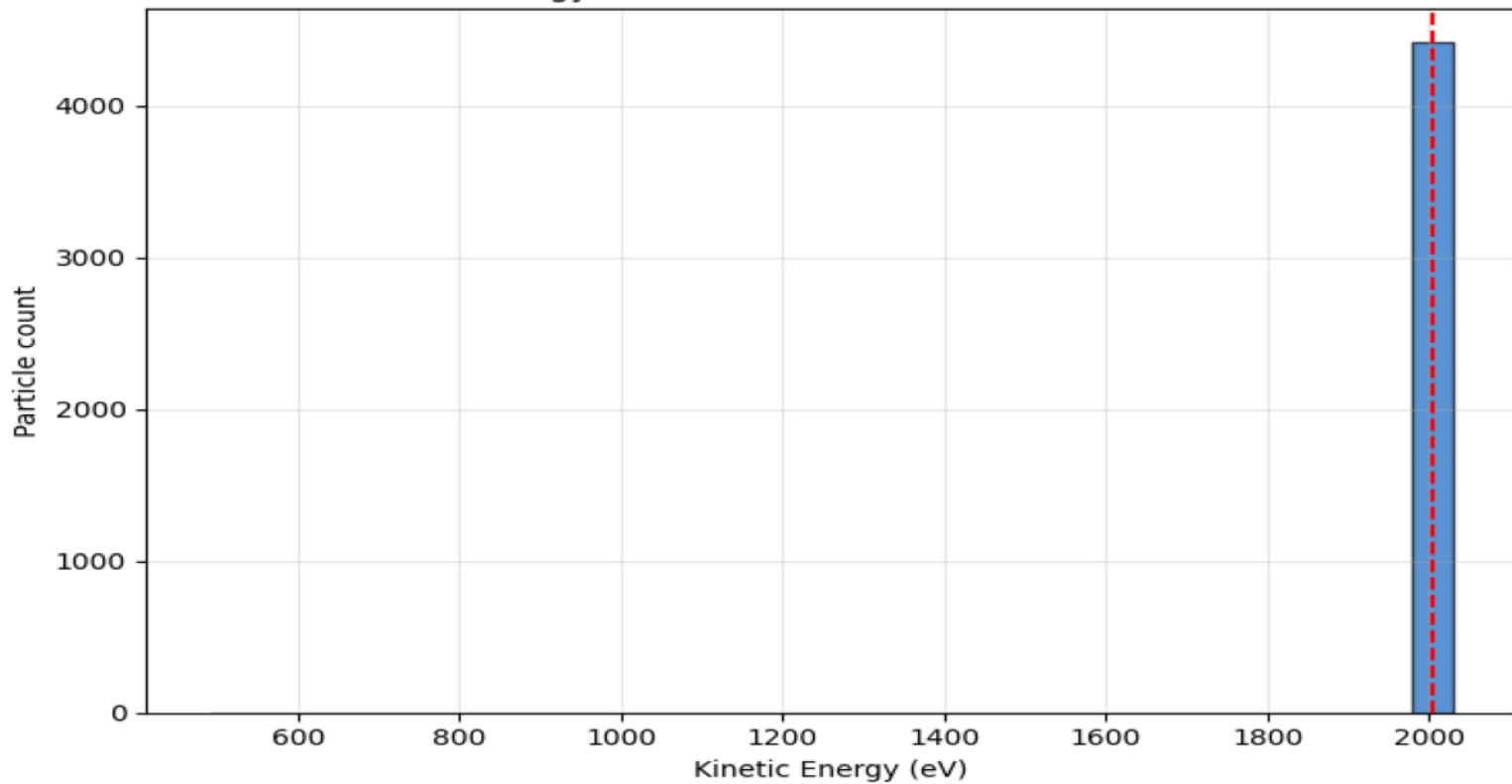
Convergence

Convergence History



Energy Distribution

Energy at x=4.9mm $E=2003.8 \pm 23.10$ eV



Perveance Scan

Perveance Scan (Voltage)

